

- 4 (a) Define strain.

.....
..... [1]

- (b) A copper wire of length 4.0 m has a uniform cross-sectional area of $4.5 \times 10^{-7} \text{ m}^2$.

A tensile force of 18 N is applied to the wire. This causes the wire to extend by 1.4 mm up to its limit of proportionality.

- (i) Calculate the Young modulus of the wire.

Young modulus =Pa [3]

- (ii) On Fig. 4.1, draw a line to show how the stress varies with the strain for the wire up to its limit of proportionality.

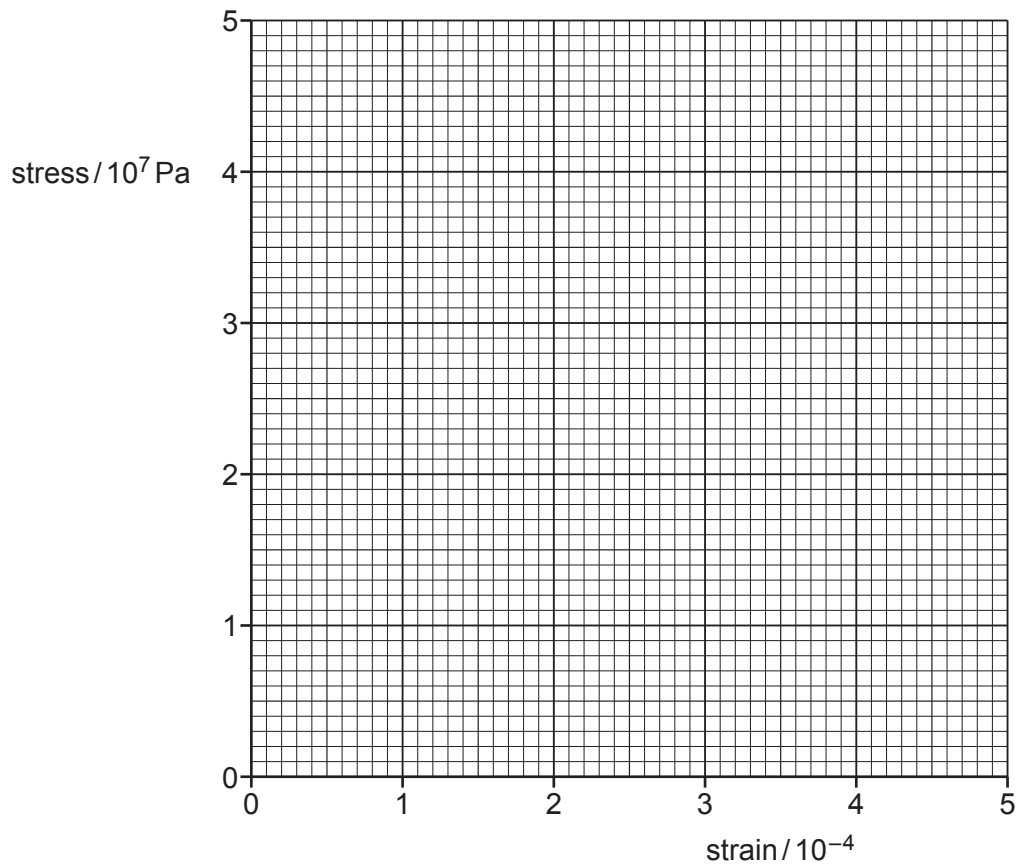


Fig. 4.1

[2]

- (c) A second copper wire has the same length as the wire in (b) but a larger diameter. Both wires are subjected to a tensile force of 18 N.

By placing a tick (✓) in each row, complete Table 4.1 to compare the stress and strain of the two wires.

Table 4.1

	greater in second wire	less in second wire	the same in both wires
stress			
strain			

[2]

[Total: 8]